

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

B.Tech. (CE/EE/IT/ CL/EC/ME) SEM - II Examination, May '11
ME - 101 Engineering Graphics

Date: 21.05.2011, Saturday

Time: 1:30p.m to 4:30 p.m

Maximum Marks: 70

- Instructions: (i) Attempt all the questions.
(ii) Figures to the right indicate full marks.
(iii) Make suitable assumptions if required.

SECTION 1

- 1.(a) I. Construct a scale of RF=1/60 to read yard and feet, long enough to measure up to 5 yards. Also show 3 feet 2 yard on the scale. [3]
II. A point A is 20 mm below HP and 0.30 dm behind the VP. Draw its projections [2]

OR

- 1.(a) Explain in brief capability of AutoCAD software. [5]
1.(b) Three points A, B & P while lying along a horizontal line in order have AB = 60 mm and AP = 80 mm, while A & B are fixed points and P starts moving such a way that AP + BP remains always constant and when they form isosceles triangle, AP = BP = 50 mm. Draw the path traced out by the point P from the commencement of its motion back to its initial position and name the path of P. [8]
2. The distance between the end projectors of a straight line AB is 80mm. Point A is 10mm above H.P. and 30mm in front of V.P. Point B is 40mm above H.P. and 50mm behind V.P. Draw the projections and find the inclination of straight line AB with H.P & V.P. and the true length of the line. [10]

OR

2. A room is 5m X 4.5m X 4 m high. Determine by method of projections of straight lines, distance between diagonally (solid) opposite corners of the room.
3. Fig. 1 shows pictorial view of an object. Draw [12]
(1) Front view in the direction 'X', (2) Left hand side view and
(3) Top view.

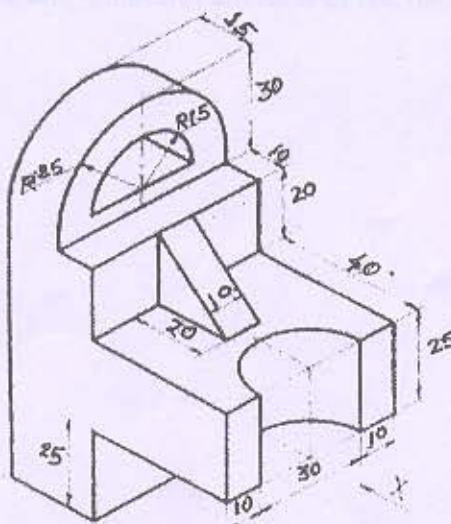


Fig 1

[P.T.O.]

SECTION 2

- 4 (a) A square pyramid of 50 mm side of the base and 50 mm length of axis is resting on one of its triangular faces on the HP having a slant edge containing that face parallel to the VP. Draw the projections of the pyramid. [8]

- 4(b) A semi-circular plate of 80 mm diameter has its straight edge in the VP and inclined at 45° to the HP. The surface of the plate makes an angle of 30° with the VP. Draw its projections. [7]

OR

- 4 (a) Draw the projections of a regular pentagon of 40 mm side, having its surface inclined at 30° to the HP and a side parallel to the HP and inclined at an angle of 60° to the VP. [7]

- 4(b) A tetrahedron of 30 mm side is resting with one of its edges on H.P. The edge on which it rests is inclined at 45° to V.P, and a face containing that edge is inclined 30° to H.P. Draw the projections of solid [8]

- 5 A Cylinder, diameter of the base is 43 mm and height is 58 mm is resting on HP on its base. It is cut by an AIP in such a way that the true shape of the section is ellipse with major axis is 60 mm and minor axis is 43 mm. find the inclination of AIP with the HP and draw three projections [10]

OR

- 5 A square pyramid, side of base 50mm and height 50 mm, is resting on H.P. on its base with the entire base equally inclined to V.P. A rectangular slot, 40 mm wide and 15 mm high, is made in the centre at the bottom of the pyramid. Draw the development of the lateral surfaces of the pyramid and top view. [10]

- 6 A frustum of a cone having bottom surface 100 mm diameter and upper surface 50 mm diameter. The height of frustum is 80mm. Draw the isometric view when the axis of that is parallel to both the reference plane [10]